Image Creation

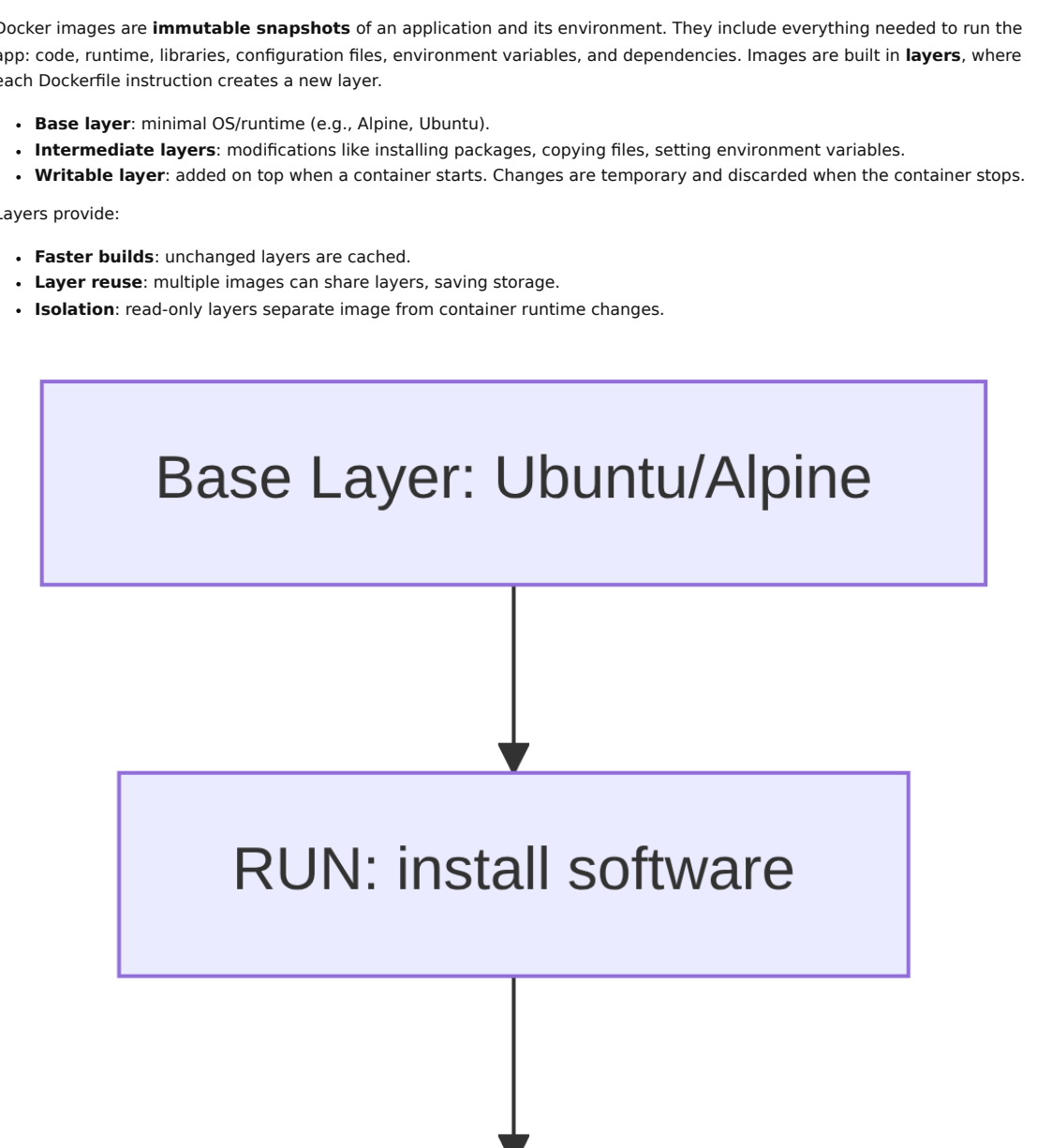
Docker Images and Layers

Docker images are **immutable snapshots** of an application and its environment. They include everything needed to run the app: code, runtime, libraries, configuration files, environment variables, and dependencies. Images are built in **layers**, where each Dockerfile instruction creates a new layer.

- **Base layer**: minimal OS/runtime (e.g., Alpine, Ubuntu).
- **Intermediate layers**: modifications like installing packages, copying files, setting environment variables.
- **Writable layer**: added on top when a container starts. Changes are temporary and discarded when the container stops.

Layers provide:

- **Faster builds**: unchanged layers are cached.
- **Layer reuse**: multiple images can share layers, saving storage.
- **Isolation**: read-only layers separate image from container runtime changes.



You can inspect layers with tools like dive.

Key Dockerfile Instructions

- `FROM <image>` : sets the base image.
- `ARG <var>` : defines build-time variables.
- `COPY <src> <dest>` : copies files or directories into the image.
- `RUN <cmd>` : executes commands during image build.
- `CMD ["cmd"]` / `CMD-SHELL "cmd"` : sets default container command. Only the last `CMD` is effective.
- `ENTRYPOINT ["cmd"]` : sets a fixed container process.
- `WORKDIR <dir>` : sets the working directory for subsequent commands.
- `EXPOSE <port>` : documents the port the app will listen on (does not publish it).
- `ENV <key>=<value>` : sets environment variables inside the image.
- `USER <user>` : sets the user for subsequent commands in the container.

> ***Difference between `RUN` and `CMD`**: `RUN` executes at **build time**, `CMD` at **container start**.*

Example: Python Application

Python script

```
# app.py
print("Hello, Docker!")
```

Dockerfile

```
FROM python:3.9-slim
WORKDIR /app
COPY . .
CMD ["python", "app.py"]
```

Build and Run

```
docker build -t $(basename "$PWD"):latest .
docker run simple-app-python
```

Example: Java Application

Java code

```
// App.java
public class App {
    public static void main(String[] args) {
        System.out.println("Hello, Docker!");
    }
}
```

Dockerfile

```
FROM eclipse-temurin:21
WORKDIR /app
COPY . .
RUN javac App.java
CMD ["java", "App"]
```

Simple Command Image

Dockerfile for `ls` command:

```
FROM alpine:3.18
CMD ["ls", "-al"]
```

Flask Echo Server

```
FROM python:3.9-slim
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

```
docker build -t $(basename "$PWD"):latest .
docker run -p 5000:5000 echo-server-flask
```

Test with:

```
curl -X POST http://localhost:5000/echo -H "Content-Type: application/json" -d '{"message": "Hello"}'
```

Spring Boot Echo Server

```
FROM eclipse-temurin:21
ARG JAR_FILE=target/*.jar
COPY ${JAR_FILE} application.jar
ENTRYPOINT ["java","-jar","/application.jar"]
```

```
mvn clean package -Dmaven.test.skip=true
docker build -t $(basename "$PWD"):latest .
docker run -p 5000:5000 echo-server-java
```